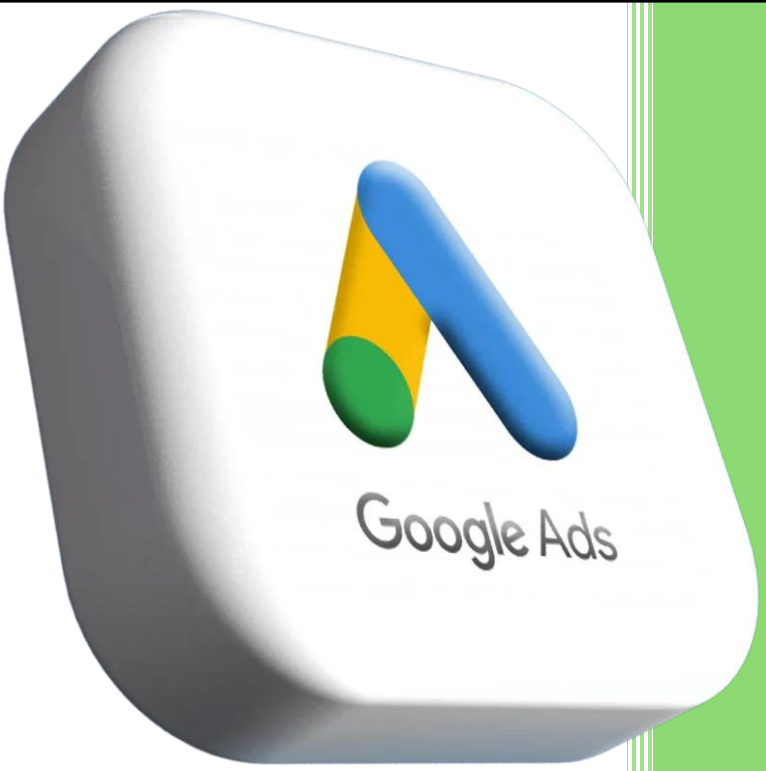
Google ads | performance-analysis



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Introduction

This report presents the results of an analysis conducted on the performance of a retail advertisement for a shop primarily selling desktops and mobile devices. The advertising campaign is run through Google Ads on a monthly paid basis. Each record includes essential marketing performance metrics such as impressions, clicks, conversions, cost, revenue, and profit or loss.

The main purpose of this analysis is to understand the patterns associated with many campaign dimensions, including **device, promotion type, match type, and time**. Moreover, it aims to explain why some campaigns are profitable while others are not. The report will conclude with a campaign strategy developed based on the insights gained from this analysis.

This report is crucial for business owners interested in promoting their businesses through digital marketing. It will show that many campaigns can generate high traffic, such as clicks, impressions and CTR. But maybe they do not lead to increased profits. Actually, they may even result in a loss of profit. To tackle this issue, the analysis will also include financial performance indicators, such as profit and loss statements, revenue, and efficiency metrics.

Project Objectives

- Assess the effectiveness of advertising campaigns using performance indicators such as **Revenue, P&L, CTR, and Conversion Rate**.
- Classify which campaign led to **higher sales volume versus higher profitability**.
- Explore campaign performance across **devices, promotion types, match types, and months**.
- Study how marketing motivations such as **coupons and promo codes** influence customer behavior.
- Detect patterns that indicate inefficient advertising costs.
- Provide **data-driven recommendations** to improve campaign strategy

Dataset Overview

The dataset contains campaign-level performance metrics across multiple months. Each row represents the performance of a campaign segment within a specific month.

Key variables in the dataset include:

- **Ad Group** – Campaign targeting strategy
- **Month** – Time period of the campaign
- **Impressions** – Number of times the advertisement was displayed
- **Clicks** – Number of users who clicked the advertisement
- **CTR (Click-Through Rate)** – Clicks divided by impressions
- **Conversions** – Number of users who completed a purchase or desired action
- **Conversion Rate** – Conversions divided by clicks
- **Cost** – Total advertising spend
- **CPC (Cost per Click)** – Cost divided by number of clicks
- **Revenue** – Revenue generated from the campaign
- **Sale Amount** – Total value of products sold
- **P&L (Profit & Loss)** – Net profitability after subtracting advertising cost

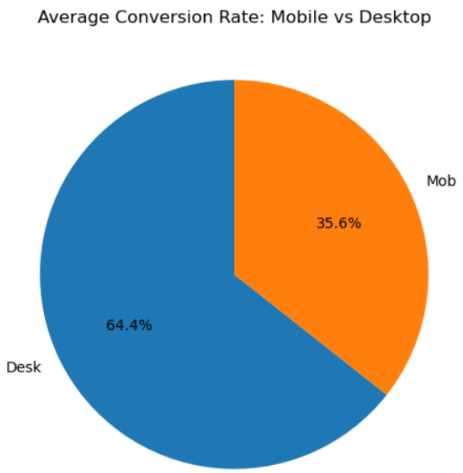
The dataset does not contain missing values and is suitable for direct analysis.

Exploratory Data Analysis

Device Performance Analysis

The number of advertising groups was nearly distributed across mobile and desktop devices, meaning that there will be a fair comparison between the two sectors.

Mobile campaigns generated **53% of total sales**, indicating that mobile device campaigns have a major effect on the traffic of purchases. However, mobile campaigns produced larger financial losses compared to desktop campaigns.

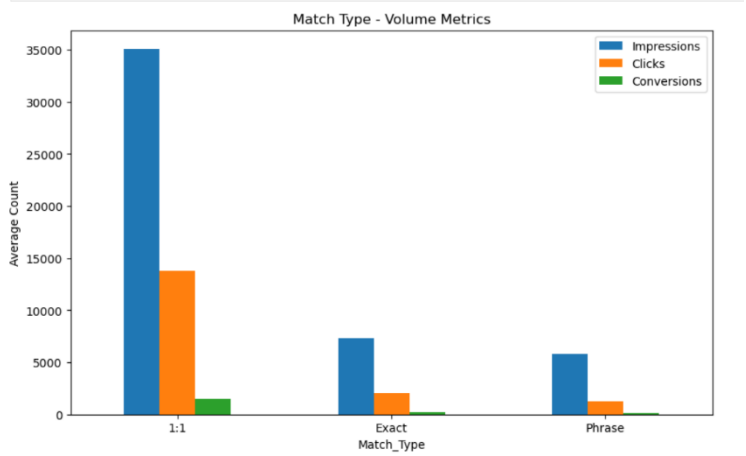


It is suggested that mobile campaigns are successful at attracting people and generating higher clicks, but the people who already convert to making a purchase are relatively few compared to the people who click, generating higher costs for streaming mobile campaigns. This assumption is proven by the following figure.

This result indicates that Mobile users are more likely to browse rather than make an actual purchase. Implementing **post-click incentives such as promotional pop-ups or discount codes** could help convert these users and improve the profitability of mobile campaigns.

Match Type Analysis

The dataset contains different keyword match strategies, including 1:1 campaign structures, phrase match, and exact match campaigns. The ad group column is separated due to these match types to study the influence of each one alone.



The **1:1 campaign structure** generated the highest number of impressions and clicks due to its working strategy, where each keyword is associated with a specific advertisement. As a result, it reached a wider range of search traffic, improving the visibility in search results.

Impressions and clicks for Exact and Phrase match types have decreased in criticality compared to 1:1, because they target narrower ranges.

Exact match campaigns are the most restrictive since ads are triggered only when the search query closely matches the keyword. Phrase match campaigns are slightly broader, allowing ads to appear for searches that include the keyword phrase in a larger query.

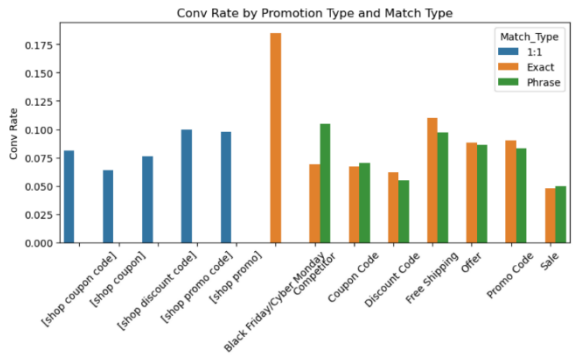
However, a 1:1 match type results in the lowest P&L, indicating that the campaign is widely streamed to people who aren’t interested in purchasing, or the keyword used in their search didn’t match the campaign's offer. This will be shown in the promotion type analysis.

Promotion Type Analysis

Promotion strategy plays a major role in influencing user behavior and campaign performance. The dataset includes several promotion types, including **coupon codes, discount codes, promo codes, and special event promotions such as Black Friday and Cyber Monday**. We will discuss the influence of each metric on these keyword types.

Click-Through Rate (CTR)

Promotion types containing the word **"code"**, such as *shop coupon code*, *shop discount code*, and *shop promo code*, consistently generate the highest CTR values.

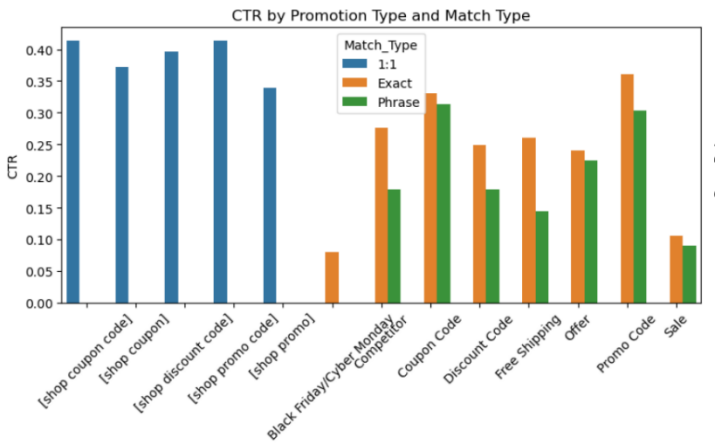


It means that users are strongly attracted to advertisements that promise discounts or promotional codes. This tells us about the users who seek any offer that saves money and widely browse through the offers.

However, a high CTR alone does not guarantee business success.

Conversion Rate

Most promotion types have similar average conversion rates. However, campaigns associated with **time-limited promotional events such as Black Friday and Cyber Monday** achieve noticeably higher conversion rates. It makes sense as customers are more motivated to convert to the offer before it is concluded.



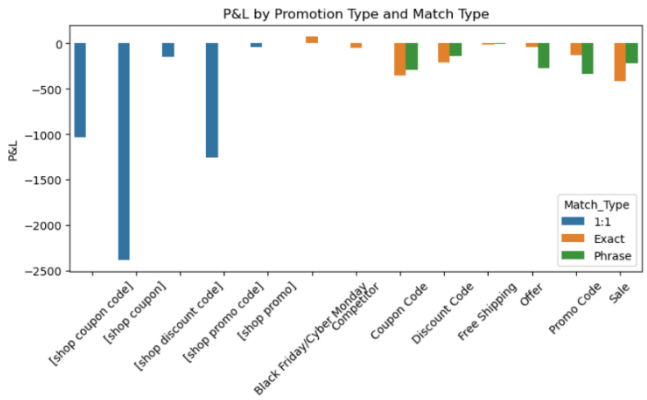
Profitability (P&L)

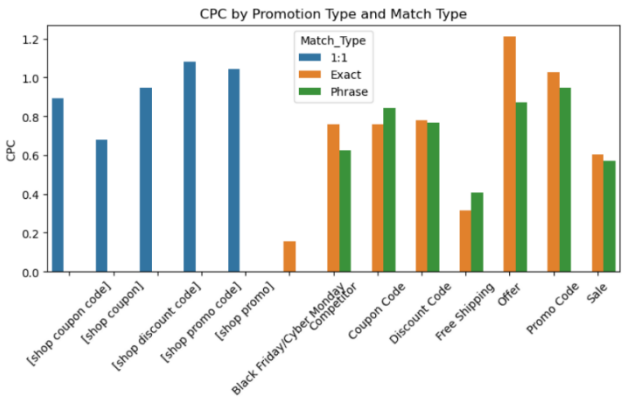
Despite their high CTR, promotion types such as **shop coupons and discount codes have the most negative impact on P&L**.

Several factors explain this pattern:

- High **Cost per Click (CPC)**
- Large numbers of clicks but relatively fewer conversions
- Heavy competition for discount-related keywords

The promotion type **"Offer"** has one of the highest CPC values in the dataset. This suggests strong competition for these keywords, which drives advertising costs upward.





As discussed before, the 1:1 strategy has the highest broad but the lowest impact on P&L. We can conclude from the CPC distribution that the word "Coupon" was associated with this match type. In addition, the CTR to these keywords was also the highest. Indicating that there is a mass search for coupons done by people who are searching for only saving money and are willing to make a purchase only for the cheapest item.

General Insights:

High CTR does not guarantee profitability.

Campaigns containing **coupon and discount codes generate large amounts of traffic but may lead to financial losses due to a low conversion rate.**

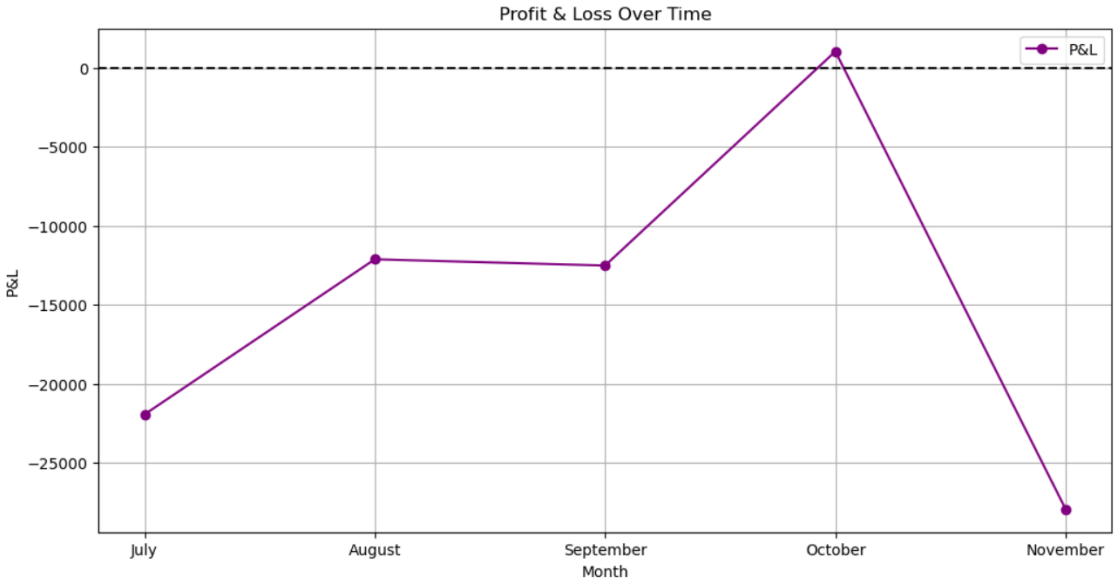
Limited promotional events such as **Black Friday or Cyber Monday** combine strong action with stronger conversion performance.

The **1:1 campaign structure generates high traffic**, while **limited offers produce strong conversions**; combining these strategies could significantly improve campaign performance.

Change Over Time Analysis

October

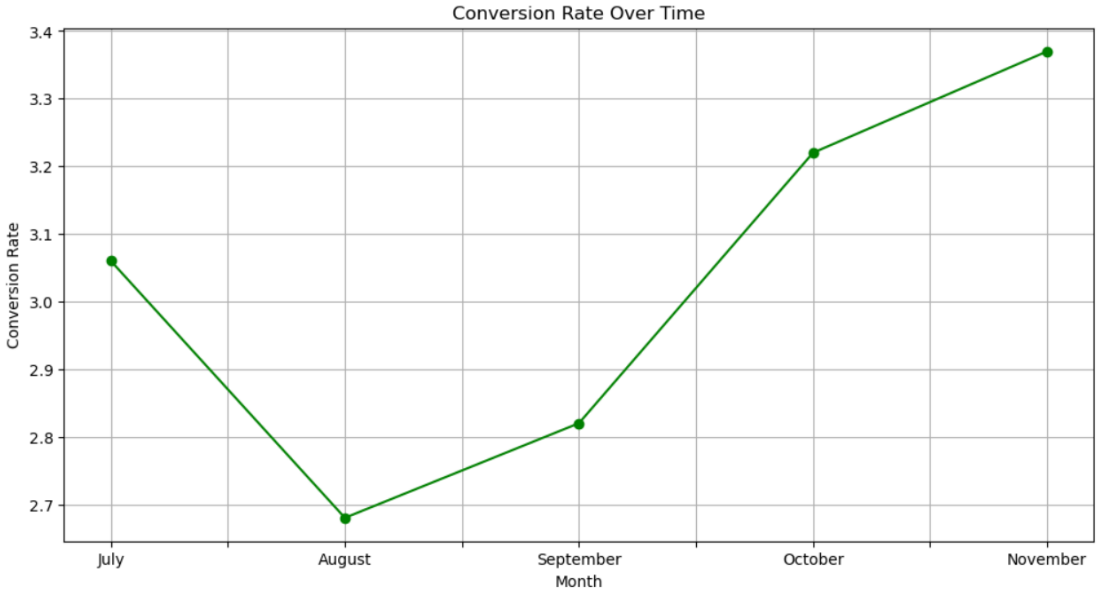
October was the only month with a **positive P&L**. During this month, the conversion rate was relatively high, and CTR remained strong, suggesting that the advertising strategy successfully attracted and converted customers.

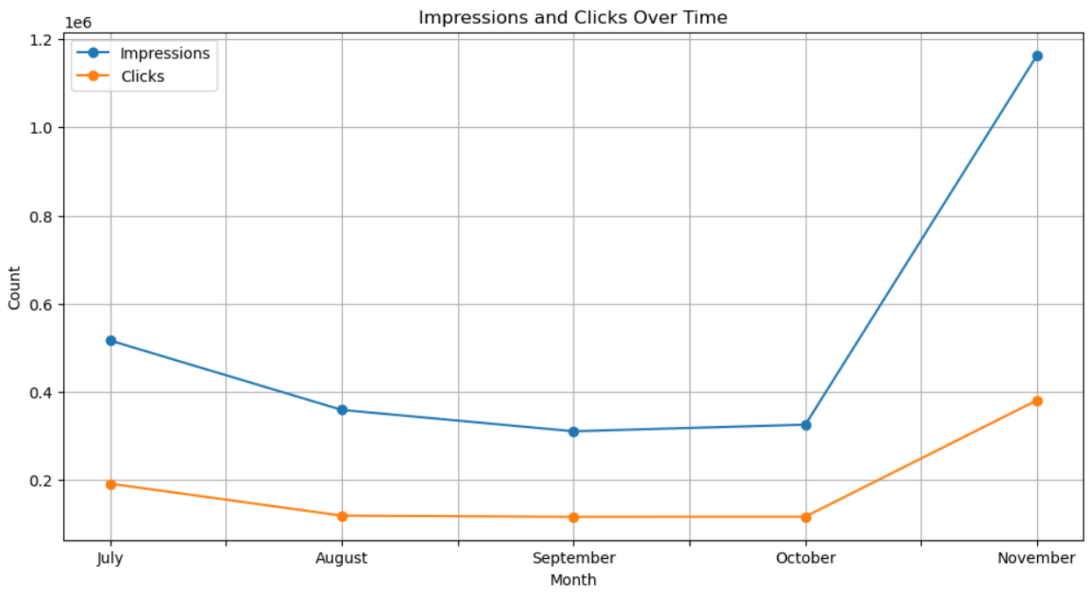


November

November had the highest conversion rate, but despite this strong performance, the month recorded the largest financial losses.

This occurred because impressions were extremely deep while the click rate remained relatively low. As a result, the campaigns generated large visibility but did not make a conversion rates.





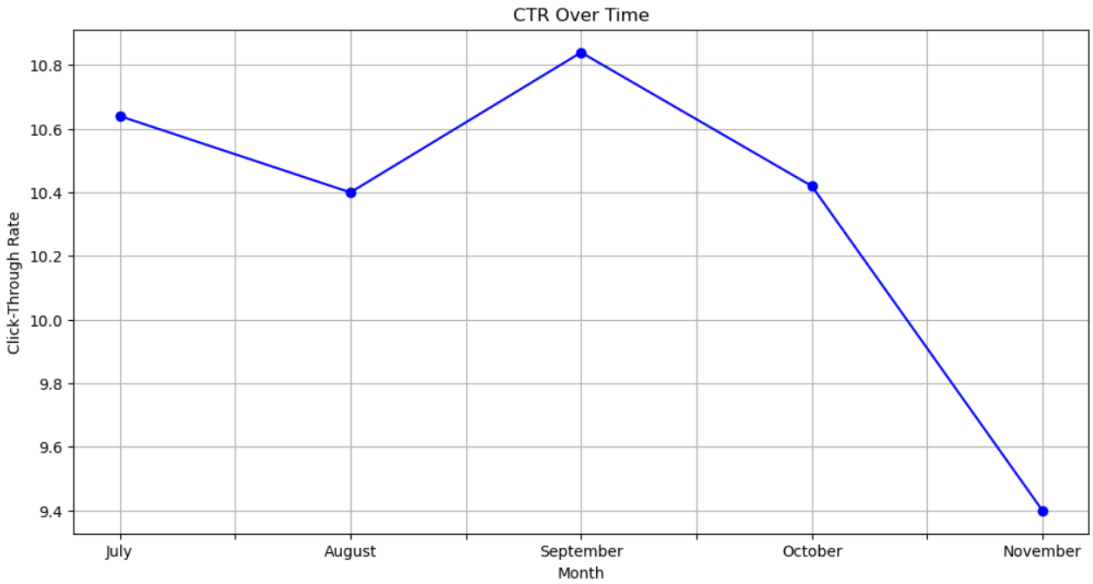
August

August recorded the **lowest conversion rate** in the dataset. CTR was relatively good; many of these interactions did not lead to purchases.

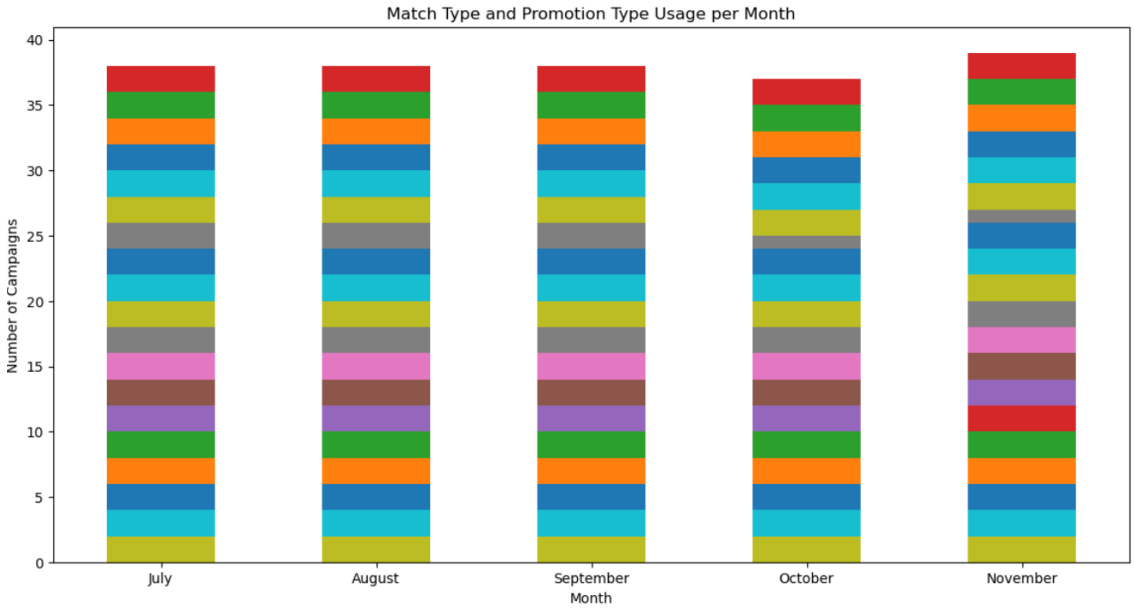
September

September produced the **highest CTR**, meaning advertisements were highly attractive to users. However, the conversion rate remained relatively modest, suggesting that while the

campaigns were effective at capturing attention, they were less successful at converting that attention into sales.



In general, the change over time in CTR, Conv rate, and P&L is due to seasonal customer behavior, as the keywords used in these months were evenly distributed, so there is no correlation between months and keywords used.



Regression analysis

Using Python, a regression model was conducted between **P&L** and all the numerical and categorical columns in the data set, and these are the key findings:

-**Coupon-based promotions have a significant negative impact on profitability**.

-In contrast, the regression analysis shows a **positive effect for promo-based campaigns**.

-Match Type (Exact/Phrase) → slight positive contribution

Insights:

-Coupon-based promotions have a negative impact, especially in customer-sensitive sectors where consumers are constantly searching for the lowest prices and discounts. This creates a group of "price hunters" who may click on various campaigns while browsing for offers but eventually do not make a purchase. These clicks can drive up the costs of campaign spending, resulting in losses that outweigh any potential gains from this segment of customers.

-Promo-based promotions have a positive impact, introducing another sector of customers who clicked on the ad with a high possibility of purchasing and only need a small incentive to finalize the transaction.

-Match type has a slight positive impact as the user search query is effectively matched with the keyword used, leading them to the right item to purchase.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               Q("P&L")   R-squared:                       0.419
Model:                            OLS   Adj. R-squared:                  0.361
Method:                 Least Squares   F-statistic:                     7.293
Date:                Fri, 13 Mar 2026   Prob (F-statistic):           2.92e-13
Time:                        22:13:56   Log-Likelihood:                -1510.6
No. Observations:                 190   AIC:                             3057.
Df Residuals:                     172   BIC:                             3116.
Df Model:                          17
Covariance Type:            nonrobust
==================================================================================================================
                                                   coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------------------------
Intercept                                      -426.6107    448.176     -0.952      0.342   -1311.245     458.023
C(Promotion_Type)[T.Competitor]                   8.1479    619.815      0.013      0.990   -1215.275    1231.571
C(Promotion_Type)[T.Coupon Code]               -349.8013    681.610     -0.513      0.608   -1695.198     995.595
C(Promotion_Type)[T.Discount Code]             -109.9715    623.534     -0.176      0.860   -1340.736    1120.793
C(Promotion_Type)[T.Free Shipping]             -108.8448    584.065     -0.186      0.852   -1261.702    1044.013
C(Promotion_Type)[T.Offer]                      -14.9631    651.147     -0.023      0.982   -1300.231    1270.305
C(Promotion_Type)[T.Promo Code]                -202.4001    700.740     -0.289      0.773   -1585.557    1180.756
C(Promotion_Type)[T.Sale]                      -217.0665    582.721     -0.373      0.710   -1367.271     933.138
C(Promotion_Type)[T.[shop coupon code]]        -452.3096    330.650     -1.368      0.173   -1104.963     200.344
C(Promotion_Type)[T.[shop coupon]]            -1844.8339    290.676     -6.347      0.000   -2418.586   -1271.082
C(Promotion_Type)[T.[shop discount code]]       476.6001    321.084      1.484      0.140    -157.172    1110.372
C(Promotion_Type)[T.[shop promo code]]         -604.7084    347.811     -1.739      0.084   -1291.236      81.819
C(Promotion_Type)[T.[shop promo]]               667.2540    297.254      2.245      0.026      80.519    1253.989
C(Match_Type)[T.Exact]                          652.2764    477.051      1.367      0.173    -289.352    1593.905
C(Match_Type)[T.Phrase]                         679.1108    530.297      1.281      0.202    -367.616    1725.838
C(Device)[T.Mob]                               -430.4786    218.426     -1.971      0.050    -861.620       0.662
CPC                                            -393.9640    302.648     -1.302      0.195    -991.347     203.419
Q("Conv Rate")                                  298.8000   1222.541      0.244      0.807   -2114.315    2711.915
CTR                                             928.9317   1355.068      0.686      0.494   -1745.771    3603.635
```

Summary findings

-Mobile campaigns generate the highest sales volume but are less profitable than desktop campaigns.

-Coupon and discount code campaigns attract strong engagement but often lead to financial losses.

-High CTR does not necessarily translate into profitability.

-Limited promotional events such as Black Friday produce stronger conversion rates.

-1:1 campaign structures generate high impressions and clicks due to their targeted keyword strategy.

-Many campaigns generate revenue but still produce negative P&L due to high advertising costs.

-Overall campaign profitability varies significantly across months.

Decision making

Optimize mobile campaign conversions

Mobile campaigns generate high engagement but a lower conversion rate. It is recommended that to enhance user experience, especially with mobile advertising, to motivate purchase. Such as improving mobile landing pages, checkout processes, and post-click incentives, like a discount.

Reduce reliance on coupon-heavy campaigns

Coupon-based campaigns attract “Price Hunters” users who may not generate sustainable profit. These campaigns should be carefully optimized or limited and concentrate on promo offers and study customers who use them because they are probably willing to convert into purchases.

Prioritize high-performing promotional events

Limited-time promotions such as Black Friday or Cyber Monday generate strong conversion performance and should receive higher advertising investment.

Combine targeted campaigns with strong promotional incentives

Since 1:1 campaigns generate large traffic volumes and limited offers produce strong conversions, combining these two strategies could significantly improve overall performance.